

When is FF at 15 Valid?

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Abstract

League of Legends is one of the largest MOBA (Multiplayer Online Battle Arena) games to this day with very few that haven't heard of it. There are two teams of five with five roles: Top, Jungle, Middle, Bottom, and Support. The teams are split into a blue side and red side and compete against each other in their respective lanes on the map Summoner's Rift. The game is won by collecting objectives, killing the other team, and destroying structures until the player eventually destroy the nexus which ends the game. These games can last anywhere from 15 minutes to 50 minutes, with the former being the earliest someone can forfeit, colloquially referred to by FF. Surrendering is viewed on a spectrum ranging from those who try to surrender for any reason and those who never vote yes to the FF. While an early forfeit can result in an avoidable loss, it can also save upwards of 15 minutes in a game that's already decided. Thus, the purpose of this experiment is to analyze key factors that contribute to a predetermined loss by the 15 minutes mark.

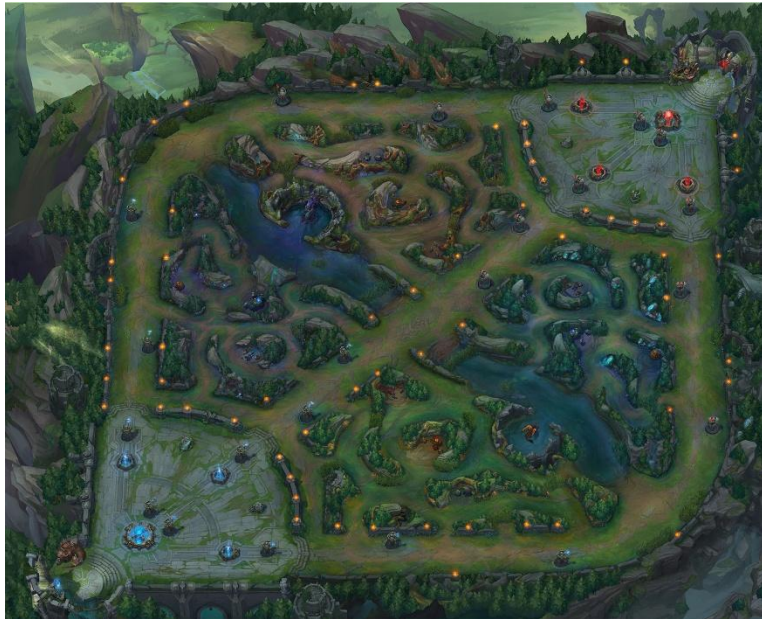


Figure 1: League of Legends Summoner's Rift

Data Description

The data was sourced from Kaggle and is data collected from approximately 20,000 diamond ranked matches in patch 14.6 from March 20th to March 31st. The dataset contains multiple metrics that can be obtained within 15 minutes of gameplay such as kills, ward counts, towers destroyed, and others. The data used in this analysis is the result of the game, gold difference, kills, and first blood. Kills are obtained when a player reduces the opponent's hit points to zero and first blood refers to very first kill of the game. First blood awards 400 gold, but as a standard, each kill awards a base 300 gold and can vary based on how many consecutive kills or deaths a player has. The more kills you have, the higher your bounty will be when you are slain with a cap of 700 additional gold; while consecutive deaths lower your bounty to a minimum of 100. Getting a kill resets the bounty to 300 if it was below and, in most cases, dying resets it as well if it was above the base amount. Getting an assist, or helping during a kill, also earns gold. There are multiple sources to earn currency in this game which contributes

to the gold difference. Gold is used to buy items to get stronger, so it is one of the most important things in this game. In the dataset, each game is unique with no missing values. The metrics used are binary with 0 representing a “no” or “loss” and 1 representing a “yes” or a “win.” Kills are always positive, countable, and discrete with a maximum value of 40 and a minimum of 0, within the data. With each variable defined, we may begin the preliminary analysis.

Analysis

Preliminary Analysis

In the dataset, there are 89 columns: one unique game ID and 44 metrics for blue side and red side. I decided to center my analysis on what might contribute to a victory the most and that led me to selecting first blood, kills, and gold difference. There is data for both sides, but I decided to only analyze these factors for blue side. It is known that blue side has a slight advantage because of the map and so match making rating (MMR) tries to fix this. The way Summoner's Rift works is that there are two main objectives, dragons and baron. Baron is more open to blue side, covered by a wall for red side, and provides a temporary buff that helps end the game, but it spawns at 20 minutes. Dragons spawn as early five minutes and respawn every five minutes after being slain. This objective can be killed four times to claim a dragon soul which provides a teamwide buff for the rest of the game. This objective is more open to red side and is covered by a wall for blue side. Additionally, after the dragon soul is claimed, the elder dragon spawns and whichever team kills it gets a temporary buff that executes enemy champions below a certain health threshold. Despite this sounding better for red side, camera position and the way the map looks makes the user interface for blue side better providing an advantage outside of the game; however, this is where MMR tries to balance things out. The players on red side tend to have a higher MMR, this means that the players are, on average, better than blue side which makes up for the disadvantage. This is where the analysis begins. Historically, blue side has had a higher win rate, but I want to determine what factors make it a better decision to surrender for red side. A better player knows how to make up for these disadvantages and can beat a worse player despite the map, but other factors may lead to a more decisive loss.

League of Legends is all about momentum, so even though first blood happens only once and can happen as early as 1 minute into the game, that can snowball into an unstoppable lead. The data was imported with all columns despite having a sorting function on the website and so I had to figure out which ones were needed. As previously mentioned, all columns had binary response except for kills, but gold difference was originally the total difference in gold and not whether or not blue team was higher. From personal experience with the game and external research, it was decided that 4000 gold difference means a team has a solid lead. For others familiar with the game, this may seem like a lot, but it only equates to one lane having a full item over their opponent, and split between 5 players, this means each player has about one more component when compared to the enemy. This doesn't happen all the time, but is by no means rare, and so I decided to make 4000 gold the threshold. To do this, another column had to be added to the original data set which returned a 1 if the difference was at least 4000, and a 0 if not. There were 17,832 games where the difference was under 4000 and 3683 games over 4000. The odds of getting a game

where you're down 4000 gold by 15 minutes is about 0.206. So, one out of every five games this will happen, which is considerably frequent considering what this can lead to, thus I decided it was fair for it to remain in the analysis. After the new data was added, I formatted the counts into a 3-dimensional table, as seen in table 1, where "GD4k" stands for the gold difference above 4000 and "FB" stands for first blood. Now everything was ready to be tested, and so I moved on to the main analysis.

Result	GD4k = No		Result	GD4k = Yes	
	FB = Yes	FB = No		FB = Yes	FB = No
Win	3918	3462	Win	2458	944
Lose	4197	6255	Lose	205	76

Table 1: Contingency tables of Result vs First Blood, by Gold Difference.

Main Analysis

The main analysis began by conducting Cochran-Mantel-Haenszel test with each variable by assuming their independence. In each case, the null hypothesis was rejected with strong evidence against it, and so my initial idea of collapsing the table was not possible. Although the tables were unable to be analyzed over the marginals, the Cochran Mantel test suggested that there is at least some dependence and so I decided to test how related each factor is compared to result with logistic regression.

The first test only used "FB" with values as seen in Figure 2. Using the values from the model, we can see that the odds of winning with only first blood is 1.44, which equates to a 59.16% chance of winning. If we assume both teams have an equal opportunity to win, obtaining the first kill of the game creates an 18% difference, since red side now had a 41% chance. The momentum gained from this can last throughout the whole game, but there are so many factors that go into the rest of the game, and first blood usually only affects one player, thus it is not enough to warrant a surrender. That is why I decided to test for a gold differential which is an accumulated lead throughout the whole 15 minutes of gameplay.

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Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.36249    0.01962  -18.48  <2e-16 ***
FB           0.73297    0.02773   26.43  <2e-16 ***

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Figure 2: Coefficient estimates from logistic regression of game result on first blood.

As seen from the values in Figure 3, with a gold difference of 4000, the odds of winning increase by 12 times with a probability of 92.37%. With the notion that red side had an innately higher chance of winning, but by being down essentially one item, the chance of winning almost completely disappears in favor of blue side. In a normal game, this would imply that red side is being outperformed by blue side, thus there are very little things that can happen if the blue team keeps up that level of gameplay. Of course, that isn't always the case, but a difference this big is hard to lose if played correctly.

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Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -0.3480    0.0152  -22.89  <2e-16 ***
GD4k          2.8418    0.0639   44.47  <2e-16 ***

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Figure 3: Coefficient estimates from logistic regression of game result on gold difference at 15 minutes.

What's more interesting is the model shown in Figure 4 with both factors applied. The probability of winning increases to 93.41%, which is only 1.04% more from the last model. When first blood isn't a factor, the probability becomes 89.64%. This means that despite first blood providing a solid lead, it doesn't contribute as much to the probability of winning compared to gold difference. This is interesting because beyond the game, losing first blood puts you behind, letting the opponent gain a steady increasing lead over you. Much of this gold difference likely stems from kills, since they're one of the fastest ways to accumulate gold early on.

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Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.57824    0.02087  -27.70  <2e-16 ***
FB           0.49480    0.02984   16.58  <2e-16 ***
GD4k         2.73596    0.06423   42.60  <2e-16 ***

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Figure 4: Coefficient estimates from logistic regression of game result on gold difference and first blood

This brings up a natural question: how many early game kills would make a surrender valid? The difference in a team's kills is most impactful early in the game, since being 7 to 15 is worse than being 30 to 38, so directly modeling the kill difference between the blue and red team on results shows how early momentum influences the outcome. Additionally, total kills are included in the model under the idea that a higher kill game reduces the weight of the difference. While positive and negative kill differences are included in the model, a positive difference means the blue team is ahead which is the focus of the analysis. Figure 5 shows the range of probabilities of winning based on the new model. The gradient is used to highlight the change in probability based on kill advantage. Around the 15-kills mark, the probability of winning is consistently within the 80% range and only gets higher from then on. Considering the flow of the game, it is unlikely for this lead to be overturned and so the gap would only grow larger as the game progresses past 15 minutes.

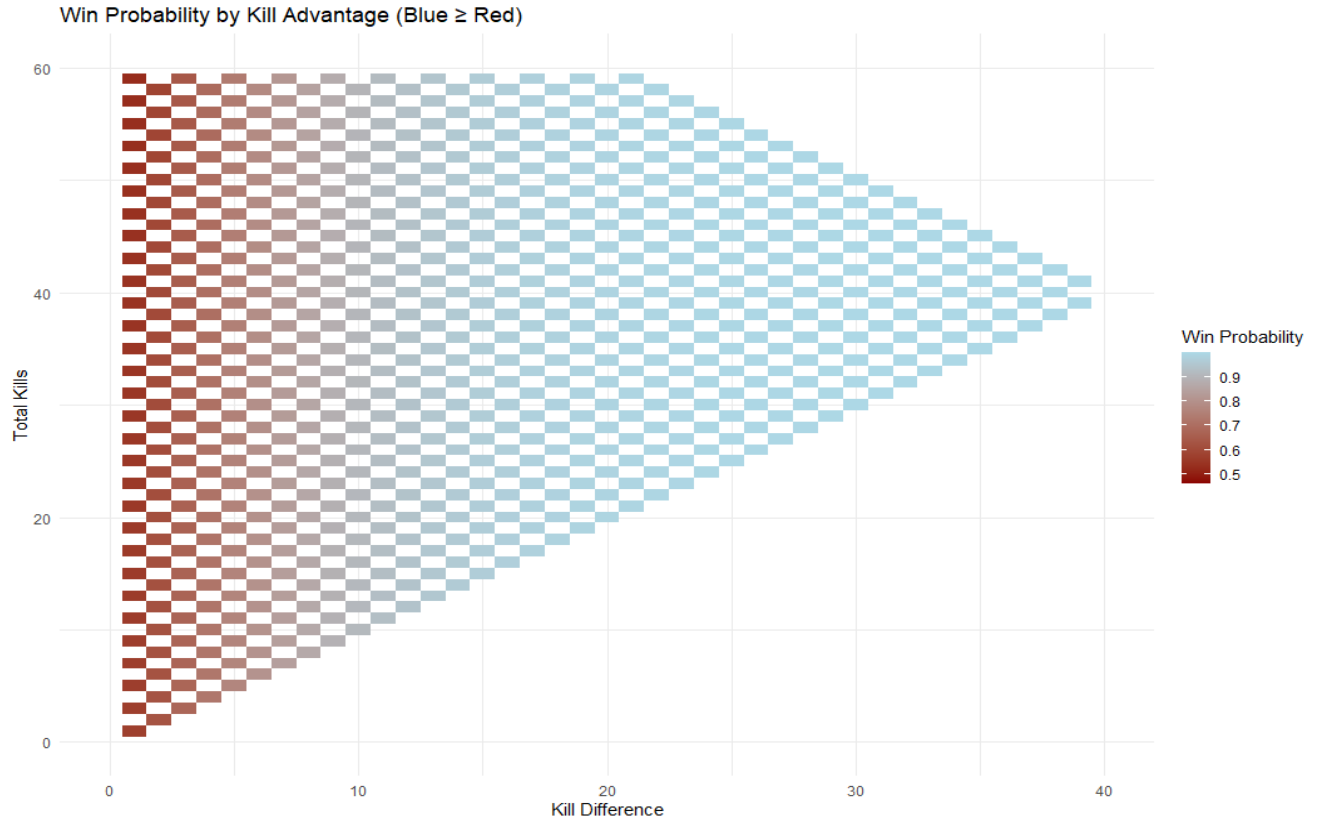


Figure 5: Plot of predicted probabilities of winning based on total kills vs kill difference using gradients.

Conclusion

League of Legends is a game of many variables that help build a lead and erase it just as quickly. The point of this analysis was to analyze the first 15 minutes of a game and model determining factors that can directly result in a victory for the blue team, hence making a surrender a reasonable option for the red team. Of the chosen variables, kill count shares a direct relationship and contributes to a good portion to their existence. Thus, it was decided to base the model on kill count, or rather the difference in kills, to be the biggest factor that results in a victory. It was discovered that when the kill difference is about 15 kills by 15 minutes, the blue team has an 80% chance of victory. This "chance" is hard to overturn considering what goes into building this lead within 15 minutes. The state of the game is a lot worse for the enemy team than the 80% losing statistic says, making it reasonable to FF at 15 when down 15 kills.